
MOLECULAR BIOLOGY (MB502T)

CURRENT FINAL TERM PAPER SPRING 2024(28July)

SHORT QUESTIONS(3)

1. Why a DNA strand with two or more AP sites gives a Poor Template ?

☐ A DNA strand with one or more AP sites makes a poor template because it lacks the information required to direct accurate replication and transcription.

2. Define Oligomers And Protomer

A multimer with just a few subunits is called as ***oligomer*** and a single subunit or a group of subunits, is called a ***protomer***. Oligomers can have either rotational symmetry or helical symmetry.

3. Translocation

The transfer of a chromosome segment to a non-homologous chromosome or a different location on the same chromosome. This results in a change in the genetic material, potentially leading to changes in gene expression, function, or regulation.

4. Function of Primer

A primer is a short single-stranded DNA sequence, typically 18-25 base pairs (bp) in length, that serves as a starting point for DNA polymerase to synthesize new strands. Primers are designed to be complementary to a specific target DNA sequence, allowing them to bind (anneal) to the target sequence and initiate DNA synthesis. Primers ensure directionality, specificity, and stability in DNA synthesis.

5. How phosphodiester bond is formed?

Phosphodiester bonds are formed during DNA and RNA synthesis through a condensation reaction between the 3' hydroxyl (-OH) group of one nucleotide and the 5' phosphate group of the adjacent nucleotide. This reaction releases a water molecule and joins the nucleotides together, forming a phosphodiester bond between the sugar and phosphate moieties of the nucleotides.

6. How and why methylation N3 in adenine is significant treatment of cancer?

Methylation at N-3 in adenine is of great practical significance because N-3 methyl adenine formation blocks DNA replication but does not appear to lead to mutations. ☐ Therefore, a methylating agent that can transfer methyl group exclusively to N-3 position in adenine would have the potential to kill cancer cells without causing mutations.

7. What is bond angle form from the rotation of C carbon in peptide bond.

☐ Pauling and Corey concluded that the peptide C – N bonds are unable to rotate freely because of their partial double-bond character. ☐ Rotation is permitted about the N - α C and the α C – C bonds. ☐ The bond angles resulting from rotations at C are labelled Φ (phi) for the N - α C bond and Ψ (psi) for the α C – C bond. ☐ In principle, Φ and Ψ can have any value between +180 & -180.

8. DNA Helicase

Primase activity is dramatically increased when it associates with another protein that acts at the replication fork called DNA Helicase. This protein unwinds the DNA at the replication fork, creating an ssDNA template that can be acted on by primase.

9. How many standard amino acids present? Name any five.

❖ Essential Amino Acids:__

Histidine (His), Isoleucine (Ile), Leucine (Leu), Lysine (Lys), Methionine (Met), Phenylalanine (Phe), Threonine (Thr), Tryptophan (Trp), Valine (Val)

❖ Non-Essential Amino Acids:

Alanine (Ala), Arginine (Arg), Asparagine (Asn), Aspartic acid (Asp), Cysteine (Cys), Glutamic acid (Glu), Glutamine (Gln), Glycine (Gly), Proline (Pro), Serine (Ser), Tyrosine (Tyr)

10. What is difference between nucleotide and nucleoside?

❖ Nucleoside:

A nucleoside is a molecule composed of a nitrogenous base (adenine, guanine, cytosine, thymine, or uracil) attached to a sugar molecule (ribose or deoxyribose).

❖ Nucleotide:

A nucleotide is a molecule composed of a nitrogenous base, a sugar molecule, and a phosphate group. Nucleotides are the building blocks of DNA/RNA.

11. Ribosome

A ribosome is a complex molecular machine that plays a crucial role in protein synthesis, also known as translation. It reads messenger RNA (mRNA) sequences and assembles amino acids into polypeptide chains, which eventually fold into proteins. Composed of two subunits (large & small), ribosomal RNA (rRNA) and proteins, ribosomes have binding sites for mRNA, transfer RNA (tRNA), and amino acids. They perform two main functions: translating mRNA into a specific sequence of amino acids and catalyzing peptide bond formation between amino acids, ensuring accurate and efficient protein synthesis. In essence, ribosomes are the "**protein factories**" of cells, responsible for creating the proteins that perform various cellular functions.

12. CRISPRs stands for ?

□ The particular stretches of unusual but characteristically organized sequence were noticed in the genomes of several bacteria which were named as of **Clustered Regularly Interspaced Short Palindromic Repeats (or CRISPRs)**. □ They consist of repeated sequences (each 30 bp long and highly conserved within a given cluster) interleaved with spacer sequences of similar length but highly divergent sequence

13. What is Plasmid & where it is found?/Why plasmids are not regarded as cell's genome.

□ Plasmids are autonomous self-replicating molecules of DNA (or very rarely RNA). □ They are not chromosomes, although they do reside inside living cells and carry genetic information. □ They are not regarded as part of the cell's genome for two reasons.

- First, a particular plasmid may be found in cells of different species and may move from one host species to another
- Second, a plasmid may sometimes be present and sometimes absent from the cells of a particular host species.

They are not needed for cell growth and division under normal conditions.

14. What is purpose of Transient Excursion in Transcription?

The "transient excursion" model proposes transient cycles of forward and reverse translocation of RNA polymerase.

Purpose: Thus, polymerase is thought to leave the promoter and translocate a short way along the DNA template, synthesizing a short transcript before aborting transcription, releasing the transcript, and returning to its original location on promoter.

15. The role of RHO protein in termination of transcriptions?

□ Rho-dependent, requires a protein called Rho to induce termination. Rho, which is a ring-shaped protein with six identical subunits, binds to single-stranded RNA as it exits the polymerase. □ The protein also has an ATPase activity, and once attached to the transcript, Rho uses the energy derived from ATP hydrolysis to induce termination.

16. RNA Polymerase in Eukaryotes

□ Bacteria have only one RNA polymerase but all **eukaryotes** have at least three different ones i.e., Pol I, II, and III; and plants also have a Pol IV and a Pol V.

17. Two features of General Transcription Factor(GTFs)?

Several initiation factors are required for efficient and promoter-specific initiation in eukaryotes. □ These are called the general transcription factors (GTFs).

- □ **In vitro**, the general transcription factors are all that are required, together with Pol II, to initiate transcription on a DNA template.
- □ **In vivo**, however, the general transcription factors are not alone sufficient to bind promoter sequences and elicit significant expression. □ Rather, additional factors are required, including DNA-binding regulatory proteins, the so-called Mediator complex, and often chromatin-modifying enzymes.

18. CTD peptides no in species

□ There are 27 of these repeats in the yeast Pol II CTD, 32 in the worm *Caenorhabditis elegans*, 45 in the fly *Drosophila*, and 52 in humans. □ The number of repeats correlate with the complexity of the genome.

19. RNA Processing Events.

Initially proteins involved in capping, then splicing factors, and finally the proteins that mediate polyadenylation.

- **Capping:** addition of a 5' cap (modified guanine nucleotide) to protect the RNA from degradation
- **Splicing:** removal of introns (non-coding regions) and joining of exons (coding regions)
- **Polyadenylation:** addition of a poly(A) tail (multiple adenine nucleotides) to the 3' end

20. What is Torpedo Model Of Termination?

□ An enzyme that degrades the second RNA as it emerges from the polymerase has been identified, and this enzyme may itself trigger termination. This is called the torpedo model of termination.

21. What is splicing?

The process of intron(Non-coding regions) removal and joining exons (coding regions) is called RNA Splicing. □ It converts the pre-mRNA into mature mRNA containing only exons.

22. Enlist the classes of splicing.

There are total three classes of splicing found in the cells:-

- Nuclear pre-mRNA
- Group II introns
- Group I introns

Thus far, we have dealt only with nuclear pre-mRNA splicing, that mediated by the spliceosome found in all eukaryotes. The so-called Group I and Group II are self-splicing introns. **Self-splicing** refers to introns that can remove themselves from RNAs in the test tube in the absence of any proteins or other RNA molecules.

23. Write two error in splicing site recognition??

- Splice-site recognition is prone to two kinds of errors.
- First, splice sites can be skipped, with components bound at, for example, a given 5' splice site pairing with those at a 3' site beyond the correct one.
- Second, other sites, close in sequence but not legitimate splice sites, could be mistakenly recognized because the splice site consensus sequences are rather loose. □ Therefore, for example, components at a given 5' splice site might pair with components bound incorrectly at such a "pseudo" 3' splice site.

24. SR Protein Work In Splicing

- So-called SR (Serine Arginine Rich) proteins bind to the sequences called exonic splicing enhancers (ESEs) within the exons. □ SR proteins bound to these sites interact with the components of the splicing machinery, recruiting them to the nearby splice sites. □ In this way, the machinery binds more efficiently to those splice sites than to incorrect sites not close to exons. □ Specifically, the SR proteins recruit the U2AF proteins to the 3' splice site and U1 snRNP to the 5' splice site. □ These factors demarcate the splice sites for the rest of the machinery to assemble correctly.
- SR proteins are essential for splicing. □ They not only ensure the accuracy and efficiency of constitutive splicing but also regulate alternative splicing.

25. Alternative splicing and its Importance

❖ DEFINITION:

□ Alternative splicing strategy enables a gene to give rise to more than one polypeptide product. □ These alternative products are called **isoforms**. □ It is estimated that 90% or more of the protein-coding genes in the human genome are spliced in alternative ways to generate more than one isoform.

OR

Many genes in higher eukaryotes encode RNAs that can be spliced in alternative ways to generate two or more different mRNAs and thus different protein products (isoforms). □ It is now believed that at least 40% of *Drosophila* genes and as many as 90% of human genes undergo alternative splicing.

❖ ROLE/IMPORTANCE:

Alternative splicing is sometimes used as a way of generating diversity, with alternative forms being generated stochastically. But in many cases, the process is regulated to ensure that different protein products are made in different cell types or in different conditions.

26. Ways of Alternative splicing?

□ Alternative splicing can occur in a number of ways. Thus, in addition to alternative exons, exons can be extended. □ In other cases, exons can be skipped (deliberately), or introns can be retained in the mature message. □ Some alternative splicing results from transcription of a gene from alternative promoters, allowing one transcript to include a 5' exon not present in the other. □ Similarly, alternative poly-A sites allow 3' terminal exons to be extended or alternative 3' terminal exons to be used in some transcripts of a given gene. There are even cases of alternative trans splicing.

27. ADAR and its types of genes

The kind of RNA editing takes place through the deamination of adenosine, which converts adenosine to inosine, which has an oxygen in place of adenine's amino group. □ **For example**, an ACG (threonine) codon becomes an ICG codon, which would be read by the ribosome as GCG (alanine).

□ This kind of RNA editing is directed by an enzyme called **adenosine deaminase acting on RNA (ADAR)**. □ Humans and mice contain three ADAR genes: ADAR1, ADAR2, and ADAR3. The products of the first two are ubiquitous in the body, but the third gene product is found only in the brain.

OR

ADAR stands for Adenosine Deaminase Acting on RNA. It is an enzyme that edits RNA by converting adenosine (A) to inosine (I), which can lead to changes in the coding sequence of genes.

28. Differentiate between prokaryotic transcription machinery and eukaryotic transcription machinery.

Prokaryotic Transcription Machinery

- Single RNA polymerase enzyme (core enzyme)
- No membrane-bound organelles (nucleus)
- Transcription and translation occur simultaneously
- No introns, genes are contiguous
- Sigma factor binds to RNA polymerase for promoter recognition

Eukaryotic Transcription Machinery

- Multiple RNA polymerases (Pol I, II, and III)
- Membrane-bound nucleus separates transcription and translation
- Transcription occurs in nucleus, translation in cytoplasm
- Introns present, genes are non-contiguous
- TFIID (TATA-binding protein) and other factors recruit RNA polymerase to promoter

29. Difference among RNA polymerase I , RNA polymerase II and RNA polymerase III

RNA Polymerase I: Transcribes ribosomal RNA (rRNA) genes, responsible for producing ribosomes.

RNA Polymerase II: Transcribes protein-coding genes into messenger RNA (mRNA), as well as some non-coding RNAs, such as snRNAs and microRNAs.

RNA Polymerase III: Transcribes small non-coding RNAs, including transfer RNA (tRNA), 5S ribosomal RNA (rRNA), and some small nuclear RNAs (snRNAs), involved in protein synthesis and RNA processing.

30. Shine Óalgarno sequence

□ To facilitate binding by a ribosome, many prokaryotic ORFs contain a short sequence upstream (on the 5' side) of the start codon called the ribosome-binding site (RBS). □ This element is also referred to as a Shine Óalgarno sequence after the scientists who discovered it by comparing the sequences of multiple mRNAs. □ The RBS, typically located 3–9 bp on the 5' side of the start codon, is complementary to a sequence located near the 3' end of one of the ribosomal RNA components, the 16S ribosomal RNA (rRNA).

31. Function of Class II release factor

Class II release factors stimulate the dissociation of the class I factors from the ribosome after release of the polypeptide chain. Prokaryotes and eukaryotes have only one class II factor called RF3 and eRF3, respectively. □ Like EF-G, IF2, and EF-Tu, class II release factors are regulated by GTP binding and hydrolysis.

FUNCTION:

- Stimulate the dissociation of Class I release factors from the ribosome after the polypeptide chain is released
- Bind to the ribosome in a manner dependent on the presence of a Class I release factor
- Remove the Class I release factor from the ribosome after it has triggered the hydrolysis of the peptidyl-tRNA linkage

32. Who are the scientists who observed that free and bound ribosomes appear to be the same, leading to the proposal of the Signal Hypothesis?

Studies by Gunter Blobel and David D. Sabitini in 1971 indicated that free and bound ribosomes appeared to be the same.

33. Why mRNA has short life span? And experiment name which analyze this statement.

mRNA has a short life span to enable cells to rapidly turn off protein synthesis when it's no longer needed, conserving energy and resources. This allows cells to quickly adapt to changing conditions and prevent unnecessary protein production, providing a regulatory mechanism for controlling gene expression and protein synthesis.

❖ Pulse-chase Experiment:

□ The particular mRNA and their life span can be analyzed by the **pulse-chase experiment**. □ Bacteria are briefly cultured in a medium that contains a radioactive precursor for RNA such as [3H] uridine. □ Then the bacteria are switched to a medium that contains a high concentration of non radioactive uridine while no [3H] uridine and samples are removed at specific times for analyses. The RNA are isolated and different types are separated by gel electrophoresis and directed by their radioactivity.

34. Two Proteins for Lactose Metabolism/Two Protein encodes by Lactose Operon /Name of two proteins in E.coli

In E. coli, two proteins are necessary for lactose metabolism. These include the enzyme β -galactosidase and a carrier molecule, lactose permease, which transports lactose (and other galactosidase) into the cell.

35. Function of β -Galactosidase

The function of the β -galactosidase in lactose metabolism is to hydrolyze lactose to form glucose and galactose. β -galactosidase also converts a small proportion of lactose to allolactose.

36. Function of GAGA binding factor

The GAGA binding factor is believed to recruit enough of the transcription machinery to the promoter for initiation of transcription. But, in the absence of a second activator, HSF, most of the initiated polymerases stall some 25–50 bp downstream from the promoter.

37. pUC Plasmid Cloning Vectors

The valuable features of pBR322 have been enhanced by the construction of a series of plasmids termed pUC (produced at the University of California) □ These plasmids are based on pBR322, from which about 40% of the DNA has been deleted. □ There is an antibiotic resistance gene for ampicillin and origin of replication for E. coli. □ In addition the most popular restriction sites are concentrated into a region termed the multiple cloning site (MCS).

38. What are Cosmids?

□ Another vector designed especially for cloning large DNA fragments is called a cosmid. □ Cosmids behave both as plasmids and as phages. □ They contain the cos sites, or cohesive ends, of λ phage DNA, which allow the DNA to be packaged into λ phage heads (hence the “cos” part of the name “cosmid”). □ They also contain a plasmid origin of replication, so they can replicate as plasmids in bacteria (hence the “mid” part of the name).

□ Cosmids are especially useful for the analysis of highly complex genomes and are an important part of various genome mapping projects.

39. Importance Of λ Phage

Bacteriophage λ is a linear double stranded phage approximately 49 kb in length. It infects E. coli with great efficiency by injecting its DNA through the cell membrane.

Phage λ has been used extensively in the production of gene libraries, mainly because of its efficient entry into the E. coli cell and the fact that larger fragments of DNA may be stably integrated. For the cloning of long DNA fragments, up to approximately 25 kb, much of the nonessential λ DNA that codes for the lysogenic life cycle is removed and replaced by the foreign DNA insert. Bacteriophages λ and M 13 are commonly used vectors and have the advantage of being able to accept large pieces (up

to 40 and 10 kb, respectively) of foreign DNA. A particularly striking and well-understood example of cooperativity in gene regulation is provided by bacteriophage λ .

40. DNA Domain

Based on the hand analogy, the three domains of the DNA polymerase are called the thumb, fingers and palm.

41. Protein Domains

Protein domains are independent units within a protein, folding and functioning on their own. They're the building blocks of protein structure, often with conserved 3D folds. Domains have a compact 3D fold, stabilized by interactions, giving them a specific structure. This structure enables specific biological roles, like enzymatic activity or ligand binding. These roles are crucial for the domain's function, allowing specific tasks within the protein. Domains perform tasks like catalyzing reactions or recognizing molecules. They're essential for protein function and interactions.

Types: Catalytic, ligand-binding, scaffolding, membrane-spanning, regulatory, etc.

42. Initiation Of Transcription

The initiation of transcription is the crucial first step in gene expression, where RNA polymerase binds to the promoter region of DNA, unwinding the double helix and positioning itself at the transcription start site. With the help of transcription factors, RNA polymerase then begins synthesizing a complementary RNA copy, either stimulating or inhibiting gene expression depending on the specific regulatory proteins involved. This tightly regulated process sets the stage for the creation of a functional RNA molecule, essential for protein synthesis, cellular regulation, and overall genetic expression.

43. RNA elongation

RNA elongation is the process by which RNA polymerase continues to add nucleotides to a growing RNA chain during transcription. After the initiation phase, where RNA polymerase binds to the promoter and begins synthesizing RNA, the elongation phase begins. During this phase, RNA polymerase moves along the DNA template, reading the base sequence and matching the incoming nucleotides to the base pairing rules (A-T and G-C). As the RNA chain grows, it is stabilized by RNA binding proteins and undergoes various modifications, such as splicing and editing. The elongation phase continues until the transcription is terminated, resulting in a complete RNA molecule that can be translated into protein or perform other cellular functions. This process is crucial for gene expression, allowing cells to synthesize specific RNA molecules in response to changing conditions.

44. Regulatory Promoter In Transcription

A regulatory promoter is a special DNA region that helps control gene activity. It's usually located near the gene and acts as a binding site for important proteins like RNA polymerase. The promoter's job is to invite RNA polymerase to start transcribing the gene, team up with other proteins to initiate or Inhibit transcription, collaborate with other regulatory elements to fine-tune transcription, and respond to external signals like hormones to adjust gene expression. There are different types of promoters, including constitutive promoters that are always "on" and regulate basic cell functions, inducible promoters that turn on in response to specific signals and regulate stress responses or metabolism, and tissue-specific promoters that are active only in certain cells or tissues and regulate development or cell specialization.

45. How D loop in secondary structure of tRNA is named ??

The D loop in the secondary structure of tRNA is named "because of the characteristic presence of dihydrouridines in the loop."

LONG QUESTIONS(5)

1. A site, P site and E site of Ribosomes and Use

The A-site is the binding site for the aminoacylated-tRNA, Delivers amino acids to the ribosome for protein synthesis. Plays a crucial role in peptide bond formation.

The P-site is the binding site for the peptidyl-tRNA, and holds the growing peptide chain in place for elongation. Participates in peptide bond formation.

The E-site is the binding site for the tRNA that is released after the growing polypeptide chain has been transferred to the aminoacyl-tRNA (E is for "exiting"). Allows for the recycling of tRNA for another round of protein synthesis.

2. Franklin and Wilkins contributions

Rosalind Franklin and Maurice Wilkins performed the x-ray diffraction analysis of DNA fibers. They showed that DNA produces a characteristic x-ray diffraction pattern.

From this pattern, they made two important findings:

- DNA molecules are helical.
- The helices have two periodicities along their long axis, a primary one of 3.4 Å and a secondary one of 34 Å

3. Explain Elongating phase of transcription?? /Role of RNA polymrase in elongation (from mid portion)

Once the RNA polymerase has synthesized a short stretch of RNA (10 bases), it shifts into the elongation phase. During elongation, the enzyme performs an impressive range of tasks in addition to the catalysis of RNA synthesis.

- It unwinds the DNA in front and re-anneals it behind.
- It dissociates the growing RNA chain from the template as it moves along.
- And it performs the proofreading functions.

Recall that during replication, in contrast, several different enzymes are required to catalyze a similar range of functions.

4. Role of RNA polymrase in elongation (from final)

During the elongation phase of transcription, RNA polymerase plays a crucial role in synthesizing and proofreading RNA. As double-stranded DNA passes through the enzyme, the strands separate at the catalytic cleft and exit via their respective channels, re-forming a double helix behind the polymerase. Ribonucleotides enter the active site and are added to the growing RNA chain, with only 8 or 9 nucleotides remaining base-paired to the DNA template at any given time. The enzyme adds one nucleotide at a time to the growing RNA transcript using a step mechanism, advancing a distance equivalent to a base pair for every nucleotide added. As polymerase escapes the promoter and initiates transcription, it sheds most of its initiation factors and recruits elongation factors like TFIIS and SPT5, as well as enzymes involved in RNA processing, to the carboxy-terminal tail of Pol II. Additionally, proteins like P-TEFb stimulate elongation by phosphorylating and activating other elongation factors like SPT5 and recruiting TAT-SF1, thereby enhancing the process. Through this complex process, RNA polymerase ensures accurate and efficient transcription of genetic information into RNA.

5. Five point of Operon model

- 1) The products of the lacZ, lacY and lacA genes are encoded in a single polycistronic lac mRNA molecule."
- 2) "The promoter for this mRNA molecule is immediately adjacent to the lac region."
- 3) "The operator is a sequence of bases (in the DNA) to which the repressor protein binds."
- 4) "When the repressor protein is bound to the operator, lac mRNA transcription can't take place."
- 5) "Inducers stimulate lac mRNA synthesis by binding to the repressor. This binding alters the repressor's conformation so it can't bind to the operator. In the presence of an inducer, therefore, the operator is unoccupied and the promoter is available for initiation of mRNA synthesis. This state is called **derepression**."

6. Which ions causes DNA damage

The ions that cause DNA damage include "hydroxyl radicals, which are produced by ionizing radiations. Hydroxyl radicals can also be produced chemically from hydrogen peroxide. Hydrogen peroxide can be converted into hydroxide radical by the following reaction: $\text{Fe}^{2+} + \text{H}_2\text{O}_2 \rightarrow \text{Fe}^{3+} + \bullet\text{OH} + \text{OH}^-$. This reaction is called the Fenton reaction and in this reaction copper, manganese and cobalt can replace iron

7. Semi Conservative Model

The Watson-Crick model proposed semiconservative replication, where the two parental strands separate and each serves as a template for a new progeny strand, resulting in one old strand and one new strand in each daughter duplex, thus conserving one parental strand in each new DNA molecule through the process of semiconservative replication. Meselson and Stahl tested this using equilibrium density gradient centrifugation, growing *E. coli* cells in a medium containing heavy nitrogen (15N) to label the DNA, then switching to light nitrogen (14N) and analyzing the DNA density in a cesium chloride (CsCl) solution, which separated the DNA molecules based on density, with 15N-DNA banding at a lower position due to its higher density. The results confirmed the semiconservative model, showing that after one generation, DNA molecules had a mix of 15N and 14N, indicating one old strand and one new strand, and after two generations, DNA molecules formed two bands: one with light nitrogen (14N) and one with a mix of 15N and 14N, demonstrating that each new DNA molecule contains one conserved parental strand and one new strand synthesized during replication.

8. Mechanism of RNA Editing

RNA editing is a process that changes the sequence of RNA molecules. It's directed by guide RNAs (gRNAs) that bind to specific regions of the RNA molecule, directing the insertion and deletion of

UMPs (uridine bases). **Editing** proceeds in a 3'→5' direction, starting at the 3' end of the RNA molecule and moving towards the 5' end. The process involves **cutting and pasting**, where the RNA molecule is cut at specific sites, and UMPs are either inserted or deleted by enzymes such as endonucleases, exonucleases, and RNA ligases. UMPs are inserted by terminal uridylyl transferase (TUTase) and deleted by an endonuclease and 3'-exonuclease. RNA ligase (for **ligation**) then seals the gaps in the RNA molecule. This process is repeated, with different successive gRNAs bind to the RNA molecule, directing editing to different regions until the entire RNA molecule is edited, allowing cells to produce different proteins from the same DNA sequence. The gRNAs act like a template, guiding the editing process to ensure that the correct changes are made to the RNA sequence.

9. Steps of attaching amino acid to tRNA

All aminoacyl-tRNA synthetases attach an amino acid to a tRNA in two enzymatic steps: ☐ Adenylation ☐ tRNA charging. Step one is adenylation in which the amino acid reacts with ATP to become adenylylated with the concomitant release of pyrophosphate. Adenylylation refers to transfer of AMP, as opposed to adenylation, which would indicate the transfer of adenine. The principal driving force for the adenylylation reaction is the subsequent hydrolysis of pyrophosphate by pyrophosphatase. As a result of adenylylation, the amino acid is attached to adenylic acid via a high-energy ester bond in which the carbonyl group of the amino acid is joined to the phosphoryl group of AMP. Step two is tRNA Charging in which the adenylylated amino acid, which remains tightly bound to the synthetase, reacts with tRNA. This reaction results in the transfer of the amino acid to the 3' end of the tRNA via the 2'- or 3'-hydroxyl and the release of AMP.

10. Changes occur in nucleosomes during transcriptions

When genes are transcribed, changes occur in nucleosomes (the basic units of chromatin) to allow the transcriptional machinery to bind to the promoter region. Two main models explain how this happens: remodeling, which reorganizes or removes nucleosomes, and modifications, which add acetyl or methyl groups to histone tails. These changes uncover DNA-binding sites, loosen chromatin structure, and create specific binding sites for proteins like bromodomains and chromodomains. The addition of acetyl groups, for example, helps the transcriptional machinery bind to nucleosomes by creating binding sites for bromodomains, like those found in the TFIID complex. This makes it easier for genes with acetylated nucleosomes to be transcribed. However, the specific requirements for these changes can vary depending on the gene and circumstances, such as the stage of the cell cycle. Important terms include: nucleosomes, remodeling, modifications, histone tails, acetyl groups, bromodomains, chromodomains, TFIID complex, and nucleosome modifiers.

11. How nucleosomes bind to promoter region

Two basic models explain how changes in nucleosomes can help the transcriptional machinery bind at the promoter. First, remodeling, and certain modifications, can uncover DNA-binding sites that would otherwise remain inaccessible within the nucleosome. For example, by removing or increasing the mobility of nucleosomes, remodelers are proposed to free up binding sites for regulators and for the transcriptional machinery. Similarly, the addition of acetyl groups to histone tails alters the interactions between those tails and adjacent nucleosomes. This modification is often said to “loosen” chromatin structure, freeing up sites. But adding acetyl groups also helps binding of the transcriptional machinery (and other proteins) in another way: it creates specific binding sites on nucleosomes for proteins bearing so-called bromodomains. One component of the TFIID complex bears bromodomains and thus binds to acetylated nucleosomes better than to unacetylated nucleosomes. Thus, a gene bearing acetylated nucleosomes at its promoter will likely have a higher affinity for the transcriptional machinery than one with unacetylated nucleosomes

12. Write initiation of replication

"The initial formation of a replication fork requires the separation of the two strands of the DNA duplex to provide the ssDNA. ssDNA is required for DNA helicase binding and to act as a template for the synthesis of both the RNA primer and new DNA. Although DNA strand separation (DNA unwinding) is most easily accomplished at chromosome ends, but DNA synthesis generally initiates at internal regions. As the circular chromosomes lack the chromosome ends so it makes internal DNA unwinding essential for the replication initiation.

The specific sites at which DNA unwinding and initiation of replication occur are called Origins of Replication. Depending on the organism, there may be as few as one or as many as thousands of origins per chromosome.

13. The Replicon Model/How the initiation of replication takes place?

It was explained by Francois Jacob, Sydney Brenner and Jacques Cuzin in 1963. This is called as The Replicon Model of replication Initiation. They defined all of the DNA replicated from a particular origin of replication as a replicon. As the single chromosome found in E. coli cells has only one origin of replication, the entire chromosome is a single replicon. In contrast, the presence of multiple origins of replication divides each eukaryotic chromosome into multiple replicons; one for each origin of replication.

The replicon model proposed two components that controlled the initiation of replication; the replicator and the initiator. The replicator is defined as the cis-acting DNA sequences that are sufficient to direct the initiation of DNA replication. This is in contrast to the origin of replication, which is the physical site on the DNA where the DNA is unwound and DNA synthesis initiates. Although the origin of replication is always part of the replicator, sometimes the origin of replication is only a fraction of the DNA sequences required to direct the initiation of replication (the replicator).

The second component of the replicon model is the initiator protein. This protein specifically recognizes a DNA element in the replicator and activates the initiation of replication. All initiator proteins select the sites that will become origins of replication. The initiator protein is the only sequence-specific DNA binding protein involved in the initiation of replication. All the remaining proteins other than initiator protein, required for replication initiation do not bind to a DNA sequence specifically.

14. Splicing Pathways

"It is striking not only that the active site is primarily formed of RNA, but also that it is only formed at this stage of spliceosome assembly. Presumably, this strategy lessens the chance of aberrant splicing."

"Linking the formation of the active site to the successful completion of earlier steps in spliceosome assembly makes it highly likely that the active site is available only at legitimate splice sites."

"Formation of the active site juxtaposes the 5' splice site of the pre-mRNA and the branch site, facilitating the first transesterification reaction."

"The second reaction, between the 5' and 3' splice sites, is aided by the U5 snRNP, which helps to bring the two exons together."

"The final step involves release of the mRNA product and the snRNPs. The snRNPs are initially still bound to the lariat, but they get recycled after rapid degradation of that piece of RNA."

15. Self-Splicing Introns

"Self-Splicing Introns reveal that RNA can catalyze RNA splicing. There are three classes of splicing found in the cells: Nuclear pre-mRNA, Group II introns, Group I introns. Thus far, we have dealt only with nuclear pre-mRNA splicing, that mediated by the spliceosome found in all eukaryotes. The so-called Group I and Group II are self-splicing introns. By 'self-splicing' we mean that the intron itself folds into a specific conformation within the precursor RNA and catalyzes the chemistry of its own release. In terms of a practical definition, 'self-splicing' refers to introns that can remove themselves from RNAs in the test tube in the absence of any proteins or other RNA molecules. The self-splicing introns are grouped into two classes on the basis of their structure and splicing mechanism. Strictly speaking, self-splicing introns are not enzymes because they mediate only one round of RNA processing. In the case of group II introns, the chemistry of splicing and the RNA intermediates produced are the same as those for nuclear pre-mRNAs.

For example, the intron uses an A residue within the branch site to attack the phosphodiester bond at the boundary between its 5' end.

16. Spliceosome

The transesterification reactions are mediated by a huge molecular "machine" called the Spliceosome. □ This complex comprises about 150 proteins and five RNAs and is similar in size to a ribosome. □ In performing even a single splicing reaction, the spliceosome hydrolyzes several molecules of ATP. □ Strikingly, it is believed that many of the functions of the spliceosome are performed by its RNA components rather than the proteins. □ Thus, RNAs locate the sequence elements at the intron – exon borders and likely participate in catalysis of the splicing reaction itself. □ The five RNAs (U1, U2, U4, U5, and U6) are collectively called small nuclear RNAs (snRNAs). Each of these RNAs is between 100 and 300 nucleotides long in most eukaryotes and is complexed with several proteins. □ These RNA – protein complexes are called small nuclear ribonuclear proteins (snRNPs – pronounced "snurps"). □ The spliceosome is the large complex made up of these snRNPs, but the exact makeup differs at different stages of the splicing reaction. Different snRNPs come and go at different times, each performing particular functions in the reaction. □ There are also many proteins within the spliceosome that are not part of the snRNPs, and others besides that are only loosely bound to the spliceosome. □ The snRNPs have three roles in splicing: - □ They recognize the 5' splice site and the branch site; they bring those sites together as required; and they catalyze the RNA cleavage and joining reactions. □ To perform these functions, RNA–RNA, RNA–protein, and protein–protein interactions are all important. □ Let us consider some of the RNA– RNA interactions. □ These interactions operate within individual snRNPs, between different snRNPs, and between snRNPs and the premRNA.

17. Gene Expression Is Controlled by Regulatory Proteins

Gene expression is controlled by regulatory proteins, which come in two types: positive regulators (activators) and negative regulators (repressors). These proteins communicate extracellular signals to genes by binding to specific DNA sites. The process of gene expression involves several steps, starting with RNA polymerase binding to the promoter, followed by a transition to an open complex, where the DNA at the start site of transcription is unwound and the polymerase is positioned to initiate transcription. Next, promoter escape occurs, where the polymerase leaves the promoter and starts

transcribing. This is followed by the elongation phase, where the polymerase proceeds through the gene, and finally, termination, where transcription is stopped. Activators stimulate transcription by helping RNA polymerase bind to DNA, while repressors decrease or eliminate transcription by blocking polymerase binding. The most common step regulated is transcription initiation, as it's energetically efficient and easier to control. However, regulating later steps can allow for more inputs and faster response times. Regulatory proteins can work by recruiting RNA polymerase to the promoter, blocking polymerase binding, or stabilizing the open complex. For example, an activator can bind to a site on the DNA near the promoter and simultaneously interact with RNA polymerase, bringing the enzyme to the promoter, a process called recruitment. On the other hand, a repressor can bind to a site overlapping the region bound by polymerase, blocking its binding and preventing transcription. The site where a repressor binds is called an operator. Understanding these steps and mechanisms is crucial for grasping how gene expression is controlled.

18. Write the regulation of galactose of an operon

The galactose operon is a catabolite-sensitive operon that contains four structural genes: *galK*, *galT*, *galE*, and *galM*, which encode enzymes for galactose metabolism. The operon is regulated by a repressor (GalR) and cAMP-CRP. GalR binds to two operators, *galOE* and *galOI*, blocking transcription. When galactose is present, it binds to GalR, causing it to release from the operators, allowing transcription. cAMP-CRP binds to the AS region, activating transcription from promoter P1 and repressing transcription from promoter P2. The operon has two promoters, P1 and P2, which are separated by 5 bp. P1 is responsible for high-level synthesis when galactose is present, while P2 allows for low-level synthesis of epimerase even when galactose is absent. The Gal repressor is a dimer that binds to both operators, causing DNA looping. The regulation of the galactose operon is efficient due to the cooperative interaction between GalR and cAMP-CRP, allowing for both high-level and low-level synthesis of galactose enzymes depending on the cellular needs.

19. Note on CTD/What do you know about Involvement of CTD tail in transcription?

The large subunit of Pol II has a carboxy terminal domain (CTD), which is referred to as the “tail”. □ The CTD contains a series of repeats of the heptapeptide sequence: Tyr-Ser-Pro-Thr-Ser-Pro-Ser. There are 27 of these repeats in the yeast Pol II CTD, 32 in the worm *Caenorhabditis elegans*, 45 in the fly *Drosophila*, and 52 in humans.

□ The number of repeats correlate with the complexity of the genome. □ Each repeat contains sites for phosphorylation by specific kinases, including one that is a subunit of TFIIF. The form of Pol II recruited to the promoter initially contains a largely unphosphorylated tail, but the species found in the elongation complex bears multiple phosphoryl groups on its tail. Addition of these phosphates helps polymerase shed most of the general transcription factors used for initiation, and which the enzyme leaves behind as it escapes the promoter. Regulating the phosphorylation state of the CTD of Pol II controls subsequent steps—elongation and even processing of the RNA— as well. Indeed, in addition to TFIIF, several other kinases have been identified that act on the CTD, as well as a number of phosphatases that remove the phosphates added by those kinases.

20. Class II Releases factors.

Class II release factors stimulate the dissociation of the class I factors from the ribosome after release of the polypeptide chain. Prokaryotes and eukaryotes have only one class II factor called RF3 and eRF3, respectively. □ Like EF-G, IF2, and EF-Tu, class II release factors are regulated by GTP binding and hydrolysis. class I release factors functionally mimic a tRNA, having a peptide anticodon that interacts with the stop codon and a GGQ motif that reaches into the peptidyl transferase center. Once the class I release factor has triggered the hydrolysis of the peptidyl tRNA linkage, it must be removed from the ribosome. This step is stimulated by the class II release factor, RF3. □ RF3 is a GTP-binding protein but, unlike the other GTP-binding proteins involved in translation, this factor has a higher affinity for GDP than GTP. Thus, free RF3 is predominantly in the GDP-bound form. □ RF3-GDP binds to the ribosome in a manner that depends on the presence of a class I release factor. After the class I release factor stimulates polypeptide release, a change in the conformation of the ribosome and the class I release factor stimulates RF3 to exchange its bound GDP for a GTP. The binding of GTP to RF3 leads to the formation of a high affinity interaction with the ribosome that favors the rotated hybrid state. □ This change in conformation displaces the class I factor from the ribosome. These changes also allow RF3 to associate with the factor-binding center of the large subunit. As with other GTP binding proteins involved in translation, this interaction stimulates the hydrolysis of GTP. In the absence of a bound class I factor, the resulting RF3.GDP has a low affinity for the ribosome and is released.

21. Three discoveries in Molecular biology.

□ In 1957, Francis Crick laid out the “Central dogma of molecular biology” which foretold the relationship between DNA, RNA, and proteins.

□ In 1958, Mathew Meselson & Franklin Stahl proved that DNA replication was semiconservative.

□ Marshall Nirenberg and Gobind Khorana working independently cracked the code in the early 1960s.

□ They found that 3 bases constitute a code word, called a codon, that stands for one amino acid.

22. Types of RNA.

There are mainly three types of Ribonucleic acids (RNAs) present in the cells of living organisms.

- **Messenger RNA (mRNA)** : It is the type of RNA that carries genetic information from DNA to the protein biosynthetic machinery of the ribosome. □ It provides the templates that specify amino acid sequences in polypeptide chains. □ The process of forming mRNA on a DNA template is known as transcription. □ It may be monocistronic or polycistronic. □ The length of mRNA molecules is variable and it depends on the length of gene.
- **Transfer RNA (tRNA)** □ Transfer RNAs serve as adapter molecules in the process of protein synthesis. □ They are covalently linked to an amino acid at one end. □ They pair with the mRNA in such a way that amino acids are joined to a growing polypeptide in the correct sequence.
- **Ribosomal RNA (rRNA)** □ Ribosomal RNAs are components of ribosomes. □ rRNA is a predominant material in the ribosomes constituting about 60% of its weight. □ It has a number of functions to perform in the ribosomes.

23. Chargaff Rule/Work of Chargaff (Late 1940s)

Erwin Chargaff and his colleagues provided a most important clue to the structure of DNA. • The work of Chargaff led him to following conclusions, also called “Chargaff Rules”:-

- 1) Base composition of DNA varies from one species to another.
- 2) The DNA isolated from different tissues of the same species have the same base composition.
- 3) The base composition of DNA in a given species does not change with an organism's age, nutritional state, or changing environment.
- 4) In DNA, the number of adenosine residues is equal to the number of thymidine (A=T) and the number of guanosine residues is equal to the number of cytidine (G=C). It means that the sum of the purine residues equals the sum of the pyrimidine residues (AG=TC).

24. Work of Watson & Crick

□ James Watson and Francis Crick postulated a three dimensional model of DNA structure in 1953. The major features of this model are as follow:-

- 1) DNA consists of two helical polynucleotide strands which are wound around the same axis to form a right handed double helix.
- 2) The hydrophilic backbones of alternating deoxyribose and phosphate groups are on the outside of the double helix, facing the surrounding water.
- 3) The nitrogenous bases of both strands are stacked inside the double helix lying perpendicular to the long axis of the helix.
- 4) The pairing and coiling of the two strands create a major groove and minor groove on the surface of the helix.
- 5) Each nucleotide base of one strand is paired in the same plane with a base of the other strand.
- 6) They also found that G pairs with C and A pairs with T due to the reason that they fit best within the structure.
- 7) The two strands in DNA are present in antiparallel orientation i.e; their 5,3phosphodiester bonds run in the opposite directions.
- 8) The two antiparallel strands of doublehelical DNA are complementary to each other.

25. Why Aflatoxin B1 can lead to replication error?

Aflatoxin B1, a powerful toxin produced by certain mold, poses a significant health risk. When ingested, it's converted into a harmful form by the body's enzymes. Normally, a small molecule called glutathione attacks and neutralizes the toxin, making it harmless. However, if the toxin escapes this attack, it can damage the DNA by binding to the guanine bases. This binding causes a distortion in the DNA structure, leading to mistakes during replication. As a result, aflatoxin B1 can cause errors during DNA replication, potentially resulting in harmful mutations. These mutations can have serious consequences, including cancer and other diseases. Therefore, it's essential to minimize exposure to aflatoxin B1, especially in food and the environment.

26. Primary structure of protein/Linear structure of protein

The primary structure of a protein refers to the sequence of amino acids that make up its polypeptide chain. This sequence is the blueprint or recipe for building a protein and determines its properties and function. Each protein has a unique sequence, like a special code, that distinguishes it from others. The sequence is written in a specific order, such as Ala-Gly-Leu-Val, and is the foundation for the higher-level structures of proteins, including the secondary, tertiary, and quaternary structures. The primary structure is the foundation for the protein's overall structure and function. It determines the protein's binding sites, enzymatic activity, and interactions with other molecules. The sequence is crucial for the protein to work properly. If the sequence is changed, the protein's function may be affected. It's the starting point for understanding how proteins work and how they're made. Primary structure ultimately determines the protein's overall shape, function, and interactions.

27. Secondary structure of proteins /Alpha helix beta helix me difference/ 1.Name two Types of chain formed as alpha and beta helix in protein.

□ The most prominent are:- Alpha-helix & β -conformations.

❖ Alpha- Helix

□ The simplest arrangement which a polypeptide chain could assume with its rigid peptide bonds is a helical structure, which Pauling and Corey called the α -helix. □ The helical twist of the α -helix found in all proteins is righthanded. □ The repeating unit is a single turn of the helix, which extends about 5.4 Å (includes 3.6 amino acid residues) along the long axis. □ The amino acid residues in an helix have conformations with $\psi = -45$ to -50 and $\phi = -60$. □ An helix makes optimal use of internal hydrogen bonds. □ About one-fourth of all amino acid residues in polypeptides are found in α -helices while in some proteins it is the predominant structure.

❖ B-Pleated Sheets

□ Pauling and Corey predicted a second type of secondary structure which they called β -sheets. □ This is a more extended conformation of polypeptide chains. □ The backbone of the polypeptide chain is extended into a zigzag structure. □ The zigzag polypeptide chains are arranged side by side to form a structure resembling a series of pleats. The R groups of adjacent amino acids protrude from the zigzag structure in opposite directions. □ Hydrogen bonds are formed between adjacent segments of polypeptide chain. □ The adjacent polypeptide chains in a sheet can be either parallel or antiparallel.

28. Quaternary structure of protein

The quaternary structure of a protein refers to the arrangement of multiple polypeptide chains, or subunits, in a protein. This structure describes how these subunits are organized in three-dimensional space to form the final protein structure. The quaternary structure includes the number and types of subunits, their arrangement in space, and the interactions between subunits, such as hydrogen bonds, ionic interactions, and disulfide bonds. Proteins with quaternary structure are called oligomers and can be classified as dimers, trimers, tetramers, pentamers, hexamers, and so on. The quaternary structure is crucial for protein function, as it can affect enzyme activity, binding affinity, and protein stability. It plays a vital role in determining the protein's overall function and its interactions with other molecules in the body.

29. Tertiary Structure Of A Protein

The tertiary structure of a protein refers to its overall three-dimensional shape, which is determined by the interactions between its amino acid sequence and the environment. This level of structure is crucial for the protein's function, as it allows the protein to bind to specific molecules, catalyze reactions, and interact with other proteins. The tertiary structure is stabilized by various forces, including hydrogen bonds, ionic interactions, van der Waals forces, and disulfide bridges. These interactions occur between different parts of the protein, such as between amino acids in different regions of the sequence or between different polypeptide chains in a multi-subunit protein. The tertiary structure can be compact and globular, like an enzyme, or more extended and fibrous, like collagen. It can also be dynamic, with different regions of the protein moving relative to each other. The tertiary structure is influenced by the protein's primary and secondary structures, as well as by environmental factors such as temperature, pH, and ionic strength. Understanding the tertiary structure of a protein is essential for understanding its function and mechanism of action.

30. Explain the mechanism of Base Excision Repair

Damage to DNA bases caused by deamination, oxidation, and alkylation is mainly repaired by one such multistep pathway which is called, **Base Excision Repair (BER)**.

❖ MECHANISM OF BER

1. Damage Recognition and Base Excision:

DNA Glycosylase Activity: The BER pathway begins with the recognition and excision of the damaged or inappropriate base by DNA glycosylases. These enzymes cleave the N-glycosyl bond between the damaged base and the sugar-phosphate backbone, creating an abasic site (AP site).

2. AP Site Processing:

AP Endonuclease Activity: The abasic site generated is processed by AP endonucleases. These enzymes hydrolyze the phosphodiester bond 5' to the AP site, creating a single-strand break with a 5'-deoxyribose phosphate (5'-dRP) and a 3'-OH group.

3. Strand Displacement Synthesis:

Depending on the type of repair needed, the pathway diverges into short patch repair and long patch repair.

- **Short Patch Repair:** In this pathway, a single nucleotide is replaced. DNA polymerase inserts a deoxyribonucleotide at the 3'-OH end using the undamaged strand as a template. Deoxyribose phosphate lyase (dRPase) removes the 5'-deoxyribose phosphate group to ensure the repair synthesis can proceed. The newly synthesized DNA strand is then sealed by DNA ligase, restoring the DNA to its intact state.
- **Long Patch Repair:** This pathway involves the replacement of 2-8 nucleotides. DNA polymerase δ or ϵ in eukaryotes extends the DNA strand, displacing the damaged segment. Flap endonuclease 1 (FEN1) removes the displaced DNA flap. Finally, DNA ligase seals the nicks, completing the repair.

OBJECTIVE

3. The Variable Loop: It sits between the anticodon loop and the U loop and, as its name varies (varies in size and sequence)

5. Hershey & Chase experiment used radioactive isotopes of phosphorus and: Sulfur.

8. 5' splice site recognized by the: U1&U6 sn RNA

9. The accumulation of toxin protein, which results from misfolding may lead to several neurological disorders except: Alzheimer's, Parkinson's, and Huntington diseases(in 3no k ilawa tha aik usy krna tha)

10. Non coding sequence are called intron

11. Nucleotide of RNA and DNA have base 5

12. The CTD contains a series of repeats of the:heptapeptide sequence.

16. Polymerase binds to DNA

17. In a signal transduction pathway, the initiating ligand is typically detected by a specific cell surface receptor: the ligand binds to an extracellular domain of the receptor, and this binding is communicated to the intracellular domain. From there, the signal is relayed to the relevant transcriptional regulator, often through a cascade of kinases.

19. Transcriptional regulatory proteins called activators help recruit polymerase to the promoter, stabilizing its binding there. ✨

1. In the second proofreading mechanism, called *hydrolytic editing* , the polymerase backtracks by one or more nucleotides and cleaves the RNA product, removing the error-containing sequence.

2. Operon proposed by *Monod and Francois Jacob*

3. Enhancers are elements that *stimulate* transcription

4. In the case of Escherichia coli, the predominant σ factor is called * σ^{70} * .

5. The Variable Loop sits between the anticodon loop and the U loop and, as its name implies, varies in size from *3 to**21 bases*

6. DNA can be best described as *double helix*

7. H3.H4 tetramer of *histones*.

8. This new RNA is recognized by an RNase called, in yeast,*Rat1.*

9. Signal transduction pathways are carried out by *Kinases* .

10. RNA polymerase can initiate a new RNA chain on a DNA template and thus *does not* *need a primer* .

11. The trp operon consists of a promoter, an operator, a leader (trpL), an attenuator, and *five* structural genes designated as trpE, trpD, trpC, trpB and trpA.

12. mRNA's short lifetime may seem wasteful, it has an important *regulatory function.*

13. Primary structure of protein also called..... ploy peptide chain)amino acids sequence

14. The small subunit contains the *decoding* center in which charged tRNAs read or " *decode* " the codon units of the mRNA.

15.The repeating units in the nucleic acids are....?

ANS:Nucleotides

16)Molecular Biology” refers to the biology of the molecules....?related to genes, gene products and heredity.

17)The site on DNA where a repressor binds is called an ...?

Ans: operator.

18. The accumulation of toxic proteins, which result from misfolding may lead to several neurological disorders including except??? *Alzheimer's, Huntington's and Parkinson's.

19) If the splice sites within the intron are too close together, splicing factors cannot bind to both sites at the same time.....?

Ans Steric Hindrance

20)The first and last codons of an ORF are known as the start and?

Ans: stop codons.

21)Enzyme that catalyzes the conversion of adenosine to....?

Ans: Inosine.

22)Chargaff work was discovered by...?

Ans:Erwin chargaff.

23)In DNA the number of adenosine residues is equal to the number of thymine?

Ans:A=T

24)The site of DNA where a repressor binds is called...?

Ans:Operator

25)Small RNA is ...!

Ans: Gene expression.

26) components of other sugar transporter systems and therefore are unlikely to be direct participants in a glucose-specific phenomenon in Catabolite repression....?

Ans: Two of these, enzyme I (E-I) and the histidine containing protein (Hpr),

27)it works by binding to site near the 3 splice site of specific exons in those pre mRNAs and recruiting the...?

Ans: U2AFsplicing factor.

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